Charting 15 years of progress in deep learning for speech emotion recognition: A replication study

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***Abstract*—Speech emotion recognition (SER) has long benefited from the adoption of deep learning methodologies. Deeper models – with more layers and more trainable parameters – are generally perceived as being 'better' by the SER community. This raises the question – *how much better* are modern-era deep neural networks compared to their earlier iterations? Beyond that, the more important question of how to move forward remains as poignant as ever. SER is far from a solved problem; therefore, identifying the most prominent avenues of future research is of paramount importance. In the present contribution, we attempt a quantification of progress in the 15 years of research beginning with the introduction of the landmark 2009 INTERSPEECH Emotion Challenge. We conduct a large scale investigation of model architectures, spanning both audio-based models that rely on speech inputs and text-baed models that rely solely on transcriptions. Our results point towards diminishing returns and a plateau after the recent introduction of transformer architectures. Moreover, we demonstrate how perceptions of progress are conditioned on the particular selection of models that are compared. Our findings have important repercussions about the state-of-the-art in SER research and the paths forward**[1].

***Index Terms*—Speech emotion recognition, Deep learning, Benchmarking, Robustness, Fairness**

I. Introduction

DEEP learning has arguably become the 'go-to' method for speech emotion recognition (SER) in the past decade. Featured prominently in recent publications and surveys [1, 2], the overwhelming consensus in the community is that deep learning (DL) outperforms previous methods. For instance, all submissions to the recent *Odyssey 2024* SER challenge relied heavily on DL models to either extract learnt representations or fine-tune them on the challenge dataset [3].

Artificial neural networks (ANNs) have been trained for SER since as early as 1998 with Huber *et al.* [4] (multilayered perceptron (MLP) for two classes), see as well Nicholson *et al.* [5] who used a shallow, 2-layer MLP in 2000. The field has come a long way since then, with one key trend being the transition from handcrafted features towards end-to-end pipelines that incorporate feature extraction and modelling within a single, monolithic architecture [6], and another being the progression towards *deeper* and *larger* methods (i. e., with more layers and more parameters) [1, 7]. Some representative examples of this trend are the early work of Trigeorgis *et al.* [6], who relied on convolutional recurrent neural networks (CRNNs) operating on raw audio, and Wagner *et al.* [7] who demonstrated the superiority of transformer models that accept as input raw audio and have been pre-trained using self-supervised learning (SSL).

However, with dozens – if not hundreds – of papers published yearly on the topics of SER and DL, it is hard to judge which advances (if any) substantially progress beyond the state-of-the-art – and why. Recent benchmark studies, such as *HEAR* [8] and *SUPERB* [9], focus only tangentially on SER, with both featuring single, small, and acted SER datasets (*CREMA-D* [10] and *IEMOCAP* [11], respectively), and only benchmarking recent DL methods. While their contribution to the general speech and audio community is undisputed, they fail to capture the progress that has been made on SER for *naturalistic* conditions. Moreover, as these studies are focused on benchmarking, they are limited with respect to their analysis of why some models perform better than others. This prevents them from uncovering important insights that can help guide future advances. For example, Wagner *et al.* [7] showed that transformer models provide substantial improvements to valence recognition, but this was later attributed to their implicit modelling of linguistic content, rather than the discovery of new, unknown paralinguistic concepts related to valence [12]. On the other hand, recurring challenge series, such as *ComParE* [2] and *Odyssey*, are more suited to capturing the *Zeitgeist* at specific moments in time, as participants compete with one another to obtain the best results using the latest methods, without necessarily revisiting older ones.

The problem is exacerbated by the fact that the field of SER is lacking a universally-accepted benchmark of speech collected in naturalistic conditions. *FAU-AIBO*, the dataset adopted by the 2009 INTERSPEECH Emotion Challenge – the world's first SER challenge – was quietly abandoned over the years with very few papers using it in its original form [13]. Similarly, the largest naturalistic SER dataset, *MSP-Podcast* [14], has seen several releases in the last few years, making comparisons

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[1]Code repository: https://github.com/CHI-TUM/ser-progress-replication

[2]http://www.compare.openaudio.eu/

across different works harder. Instead, the most widely-cited datasets became *IEMOCAP* [11] and *EmoDB* [15]. *IEMOCAP* features only acted speech (both scripted and improvised) and lacks an established evaluation scheme (with some works relying on leave-one-speaker-out cross-validation and others preferring leave-one-session-out). *EmoDB*, which is also acted, lacks a common evaluation scheme, and features only restricted linguistic content. (Note that *EmoDB* was originally intended to lay the foundations for emotional speech synthesis, not to train SER.)

As such, SER is suffering from a 'blind spot' as pertains to the progress that has been made by switching to larger and deeper models over the years. This is a blind spot we aimed to address in our recent comparison of different models on the 2009 INTERSPEECH Emotion Challenge data [13]. In the present work, we extend this study by additionally including *MSP-Podcast* as another, more recent and naturalistic, dataset of emotional speech. Furthermore, we present results for large language models (LLMs) using textual data, and supplement our performance comparison with several analyses to better understand the progress made by using DL architectures.

The remainder of our work is organised as follows. In Section II, we present relevant background, discussing benchmarking approaches and offering an overview of the most important DL trends seen in SER research in the last decade. Section III outlines our methodology and Section IV showcases our results. Section V includes a discussion of our findings and our conclusion, Section VI the limitations of our work, and Section VII our suggestions for future work.

II. Related work

This section presents related work, beginning with a discussion of benchmarking in the context of SER and continuing with a succinct overview of the main DL architecture families employed for SER.

A. Benchmarking

As discussed in Section II-B, there is a variety of architectures that have been introduced for speech analysis, and particularly for SER, but these are often evaluated on only a small number of datasets, with no consensus on common benchmarks. Typically, a single work will refer to only a few 'baseline' methods from prior work, oftentimes without reproducing the results but taking them from the literature. This makes it hard to compare different models and algorithms proposed by different groups. Crucially, it makes it hard to quantify the progress that has been made as new classes of architectures, like convolutional neural networks (CNNs) or transformers, are introduced.

The problem is further exacerbated by a gradual change in the datasets used by the majority of the community (which often, but not always, occurs for good reasons). For example, the first large-scale dataset open-sourced for SER research was *FAU-AIBO* [16], which was used for the 2009 INTERSPEECH Emotion Challenge [17]. While large (for its time) and well-known in the community, it was largely abandoned in the latter half of the 2010s with very few works using it (and even less in the form published for the challenge). Instead, the community transitioned to other datasets, such as *IEMOCAP* [11] (the top-cited SER dataset at the time of writing), *SEWA* [18], *MSP-Podcast* [14], or others. We note that this change was not always motivated by a quest for more 'naturalness' or 'bigger' data. Indeed, *FAU-AIBO* is both larger and collected in a more realistic, not explicitly prompted setting than *IEMOCAP* [11], *RAVDESS* [19], *CREMA-D* [10] and other datasets which have become more prevalent in recent years.

Moreover, the matter of properly benchmarking artificial intelligence (AI) models is far from trivial. Beyond the choice of data discussed above, it is important to define an appropriate *computational budget* and overall *engineering effort* devoted to each particular method. For example, the winner of the 2009 INTERSPEECH Emotion Challenge utilised a hierarchical task tree by identifying supergroups of target emotions using expert knowledge [20]. While this is a valid – and very innovative – approach, it is definitely more byzantine than the simple baseline which relied on 'mere' classification. Similar 'over-engineered' approaches have been proposed over the years; from ones that take into account the exact problem proposed by a dataset, to ones with elaborate feature engineering. These require substantial effort to reproduce, making their inclusion in a benchmarking effort prohibitively time-consuming. In our present work, we chose to abide by the "bitter lesson" of Sutton [21], namely, that simpler approaches will scale better given more computational resources. This fits the broader landscape of contemporary SER research – that of almost complete dominance of DL. Therefore, we focused on 'simple' deep neural network (DNN) models and trained each of them with a fixed computational budget (in the form of update steps, not compute time) and hyperparameters. While not exhaustive, and potentially disadvantaging some approaches, this allows for a more fair comparison.

B. SER model timeline

DNN architectures used for SER evolved substantially throughout the years. In the following, we present a list of the most popular SER models of recent years, beginning with the launch of the 2009 INTERSPEECH Emotion Challenge [17] and continuing to present day. As the most prominent task in computational paralinguistics (CP) research, innovation in SER has largely driven other related subfields as well, and is thus representative of the field as a whole.

'09-: *openSMILE* feature sets – In the different iterations of the ComParE Challenge, newer versions of paralinguistic features were introduced. In general, these feature sets were larger and covered a wider gamut of acoustic and prosodic features than the initial features introduced in the 2009 INTERSPEECH Emotion Challenge. However, that period saw a parallel pursuit for *smaller* expert-driven feature sets [22]. Researchers used either the *dynamic* low-level descriptors or their higher-order *static* functionals. These features were fed into MLPs or recurrent neural networks (RNNs) for further processing.

'12-: *ImageNet* pre-training – Following the introduction of *ImageNet* and the first CNNs trained on it in 2012, such networks were subsequently introduced in the audio and speech domains by substituting images with (pictorial representations of) spectrograms [23, 24] – a practice that is relevant to this day, with audio transformer models oftentimes initialised with states pre-trained on *ImageNet* [25].

'16-: End-to-end – Subsequent years saw the introduction of *end-to-end* models, i. e., models trained directly on raw audio input for the target task without any prior feature pre-processing [6]. These models were especially successful in the case of time-continuous SER, which requires predicting the emotion of very short audio frames, and usually follow the CRNN architecture.

'16-: Supervised audio pre-training – In parallel to *ImageNet* pre-training, there were also efforts to collect similar large-scale datasets for audio where networks could be pre-trained in a supervised fashion. Two notable examples are *VoxCeleb* and *AudioSet* [26], both collected from YouTube, with the former targeted to speaker identification and the latter to general audio tagging. *VoxCeleb* formed the basis for training speaker embedding models (i. e., 'x-vectors') using time-delay neural networks (TDNNs) [27]. *AudioSet* in turn inspired the use of convolutional networks, such as *CNN14*, which was introduced in pretrained audio neural networks (PANNs) [28] and shown to also transfer well to SER tasks [29], and later, transformer-based models such as *AST* [25]. In addition, the introduction of the *Whisper* architecture [30] led to a renaissance of supervised training for automatic speech recognition (ASR).

'20-: Self-supervised audio pre-training – The introduction of transformers and the advent of self-supervised pre-training for computer vision and natural language processing (NLP) also propagated to the speech and audio domain. The two dominant architectures here are *wa2v2ec2.0*(*w2v2*) [31] and *hubert* [32]. These models have yielded significant advances in SER, as also shown in our own work [7, 13], which we have partially accredited to their ability to simultaneously encode linguistic and paralinguistic information [12].

III. Methodology

This section outlines our workflow, beginning with the considered datasets, continuing with the experimental settings used for training models, and finishing with the details of the accompanying analyses.

A. Datasets and tasks

As mentioned, we employed two different datasets. See Appendix A for a broader discussion on our dataset selection criteria. *FAU-AIBO* is the official dataset of the INTERSPEECH 2009 Emotion Challenge [17]. It is a dataset of German children's speech collected in a Wizard-of-Oz scenario and annotated on the word-level for the presence of 11 emotional/communicative states by 5 expert raters [33, 34]. Subsequently, segmented words have been aggregated to meaningful chunks using manual semantic and prosodic criteria. Accordingly, annotated states have been mapped to 2- and 5-class categorisations with a set of heuristics, which forms a final dataset of 18 216 chunks used for the challenge. *FAU-AIBO*^{2} features the classes *negative* (NEG) and *non-negative* (IDL), whereas *FAU-AIBO*^{5} comprises *anger* (A), *emphatic* (E), *neutral* (N), *motherese* (P), and *rest* (R) for the remaining classes. The data is heavily imbalanced towards the neutral/non-negative classes. The data was collected from two schools, with one school set aside for testing (*Mont*) and one set aside for training (*Ohm*); we use the same partitioning for our experiments. Additionally, we create a small validation set comprising the last two speakers of the training set (speakers are denoted by number IDs): *Ohm_31* and *Ohm_32* as in our previous work [13]. Note that the data are extensively documented in S. Steidl [34].

MSP-Podcast [14] is a recently-introduced data set for SER, and one of the biggest publicly-released datasets to date. However, one 'downside' of it is that new releases are made approximately every year; these releases include increases to the data size but also corrections to the annotations and files included. We used *MSP-Podcast-v1.11* as the latest version of the dataset available to us, comprising 44 586 instances in the training set, 11 947 in the validation, and 20 845 in the test set. The dataset has been annotated for the emotional dimensions of *arousal*, *valence*, *dominance*, as well as for the emotion categories of *angry*, *contemptful*, *disgusted*, *afraid*, *happy*, *neutral*, *sad*, and *surprised*. A 7-point Likert scale was used for the emotional dimensions on the utterance level, and scores by individual annotators have been averaged to obtain a consensus vote. All versions are split into speaker independent partitions with an official train/dev/test partition; some releases feature a *test-2* partition which we ignore throughout our work.

B. Experimental settings

In total, we experimented with 43 audio-based models (see Appendix B for the complete list). All audio models were trained using *autrainer* [35]. To constrain our space of hyperparameters, we conducted two experimental phases. In the first **exploration phase**, we tested all 43 investigated models using a fixed set of hyperparameters. Specifically, we used the *Adam* optimiser with a learning rate of .0001 for 30 epochs. We used a batch size of 4 due to hardware constraints (access to higher-end GPUs that could fit all models at higher batch sizes was more limited to us than lower-end GPUs which fit all models at a batch size of 4). In total, this resulted in 43 runs for the exploration phase of each task.

Additionally, we conducted a **tuning phase**, where we further optimised a larger set of hyperparameters for the 5 best-performing models from the exploration phase for *FAU-AIBO*^{2}, *FAU-AIBO*^{5}, and *MSP-Podcast-v1.11*, doing a grid search over optimisers {*Adam*, *SGD*} and learning rates $\{.01, .001, .0001\}$, while training each configuration for 50 epochs for *FAU-AIBO* and 5 epochs for *MSP-Podcast-v1.11* (due to time constraints). For *FAU-AIBO*, we additionally explored different batch sizes $\{4, 8, 16\}$.

To account for variable-length sequences in training, we randomly cropped/padded all chunks to a fixed length of 3 seconds (different cropping/padding was applied on each instance across different epochs; random seed was fixed and can be reproduced) when using dynamic features (including the raw audio); during inference, we used the original utterances, only padding those shorter than 2 seconds with silence. Except for *w2v2* and *hubert*, where we froze the feature extractors, all model parameters were fine-tuned. We used the weighted cross-entropy loss for classification to account for class imbalance. In all cases, we used the defined validation set to select the best-performing epoch for each model, which we then evaluated on the test set.

Our search over text-based models was much more restricted. It was also inspired by recent developments in NLP, but was much more constrained in time. We evaluated *BERT* [36], *RoBERTa* [37], *DistilBERT* [38], and *Electra* [39] as "first generation" transformers, and *Llama-2* [40], *Llama-3* [41], and *Mistral* [42] as "second generation" LLMs. First generation transformers were trained with a batch size of 8, *AdamW* with a weight decay of .00001, and a learning rate of .00001 for 10 epochs. Second generation LLMs were trained for 5 epochs with a learning rate of .00001, *AdamW* with a weight decay of .01, and low-rank adaptation (LoRA) fine-tuning of only the attention projection matrices (for query, key, value, and output) [43] with 4-bit quantisation of model weights. For *FAU-AIBO*, we used the official transliteration provided by human raters. For *MSP-Podcast-v1.11*, we used automatic transcriptions generated with *Whisper-large-v2*.

C. Model evaluation

Independently and identically distributed (IID) evaluation: Models were initially evaluated on the corresponding test set of each dataset. We report unweighted average recall (UAR) – the standard metric for SER since the 2009 INTERSPEECH Emotion Challenge which accounts for class imbalance by computing first the recall per class and then averaging.

Out-of-domain (OOD) evaluation: Similar to Wagner *et al.* [7], we also evaluated OOD performance. To do so, we used the SER models trained on *MSP-Podcast-v1.11*, as this is the only task for which it was possible to find OOD data (due to the non-standard labels used in *FAU-AIBO*). Specifically, we used *EmoDB* [15] and *IEMOCAP* [11]. For both datasets, we used the entire data corresponding to the four classes that the model was trained on. For *IEMOCAP*, we additionally re-labelled samples belonging to the "excited" class as "happy". Performance was once again evaluated using UAR.

D. Probing features

In order to better understand which paralinguistic features models rely on – especially after fine-tuning – we searched for interpretable features that are predictable from the intermediate representations of transformer models. This is known as *probing* [12]. For our probing experiments, we relied on a small, easily-interpretable subset of the extended Geneva minimalistic acoustic descriptor set (*eGeMAPS*) [22]. Those were: $\underline{\mu(\text{P})}$ average pitch; $\underline{\sigma(\text{P})}$ standard deviation of pitch; $\underline{\mu(\text{L})}$ average loudness; $\underline{\mu(\text{L})\,\boxed{\text{dB}}}$ average loudness in dB; $\underline{\mu(\text{J})}$ average jitter; $\underline{\mu(\text{S})}$ average shimmer; $\underline{\mu(\text{HNR})}$ average harmonic-to-noise ratio; $\underline{\mu(\text{F1})}$ average frequency for first formant; $\underline{\mu(\text{F2})}$ average frequency for second formant; $\underline{\mu(\text{F3})}$ average frequency for third formant.

Concretely, we extracted hidden representations from all layers and all transformer models that were fine-tuned on *MSP-Podcast-v1.11*, namely, *w2v2-b*, *w2v2-L*, *w2v2-L-vox*, *w2v2-L-robust*, *w2v2-L-12-avd*, *hubert-b*, and *hubert-L*. We then averaged these representations over the time dimension. Subsequently, we used a linear regression model to predict the absolute value of each feature scaled in the range $[0, 1]$. Models were trained and evaluated using leave-one-speaker-out (LOSO) cross-validation (CV) on *EmoDB* and we measured their Pearson's r.

E. Robustness analysis

For our investigation of robustness, we created a noisy mixture of *MSP-Podcast-v1.11*, leveraging the 2021 version of the TAU Urban Audio-Visual Scenes Development dataset (*TAU-UAVS*) [44] as our source for additive noise. All audio was sourced from its *training set*, where we used the first 10 instances of each label in each city to create our mixtures. Specifically, for every test instance of *MSP-Podcast-v1.11*, we randomly drew one audio file from *TAU-UAVS* with replacement, which we added to the original audio from *MSP-Podcast-v1.11* using a signal-to-noise ratio (SNR) of $0\,\text{dB}$. We then evaluated models trained on clean audio to measure the deterioration in performance caused by additive noise.

F. Individual fairness

The standard process for computing performance in machine learning (ML) tasks is to consider each instance as an *independent trial*. Model performance is then measured using some metric, $\mathcal{M}$, over the set of reference labels $\{y\}_0^N$ and predictions $\{\hat{y}\}_0^N$, with N being the size of the evaluation set. However, the disparity in performance across different individuals plays a vital role in SER [45]. Therefore, it makes sense to consider *individual-level performance*, and, as a corollary, *individual-level fairness*.

To define individual-level fairness, we began by computing the performance on a speaker-level, thus assuming that speakers are first selected independently, and only then are samples selected independently for them (see Guyon *et al.* [46] for a similar argumentation). We finally computed the metric $\mathcal{M}$ over the set of instances for each individual speaker in our dataset, which we denote as $SP_{\mathcal{M}}$ in order to differentiate it from the standard evaluation protocol of treating each utterance independently, which we denote by $U_{\mathcal{M}}$. We call $SP_{\mathcal{M}}$ the **utility** of each speaker, as this is the benefit that each speaker can expect from getting their emotions recognised correctly[3]. We examined this utility by taking the Gini coefficient ($G(\cdot)$) as a standard measurement used in econometrics to judge the distribution of utility in a particular population; it is defined as half of the mean absolute difference relative to the mean of a particular sample [47]. The Gini coefficient has been previously used to quantify inequality in speaker-level performance for SER [45]. In particular, it takes the value of 0 for an equal distribution where everyone has the same utility, and 1 for a completely unequal one, where the entire utility is accumulated

[3]Naturally, what this 'benefit' entails depends entirely on the downstream application.

by one particular individual. In our case, this was computed as:

$$G(\{u_i\}_1^N) = \frac{\sum_{i=1}^{i=N}\left(\sum_{j=1}^{j=N}(|u_i - u_j|)\right)}{\mu(\{u_i\}_1^N)}, \quad (1)$$

where N is the number of speakers in the test set and $\{u_i\}_1^N$ is the set of utility values for all speakers, with $u_i = SP^i_{\mathcal{M}}$, i. e. , the speaker-level performance computed for each speaker.

IV. Results

This section presents our performance comparisons for all audio models both IID and OOD. Moreover, it contains our probing analysis and results on robustness and individual fairness.

A. Audio-based models

We begin the presentation of our results by exclusively considering audio-based models, i. e. , models that accept speech (either in raw form or as features). A complete list of models and pre-trained states is provided in Appendix B.

1) IID results: Results for the *exploration phase* are shown in Table I. For *FAU-AIBO*, the best-performing model for the 2-class problem was *CNN14*, with a UAR of .692, whereas for the 5-class problem it was *ResNet50*, with a UAR of .428. We further observe that some models failed to converge and yield chance (or near-chance) performance – most likely caused by the choice of hyperparameters, which favour some models more than others. Notably, most of our results were below the challenge winners for *FAU-AIBO* (UAR: .703/.417) and several were in the same 'ballpark' as the original baseline (UAR: .677/.382). For *MSP-Podcast-v1.11*, the best performance was achieved by *w2v2-L-12-avd* with a UAR of .609; however, this model has been already trained on dimensional SER on the same data by Wagner *et al.* [7], and may therefore exhibit better performance due to this cascaded IID finetuning. The second-best model was *hubert-L*, with a UAR of .570.

Is model ranking consistent across tasks? We considered model ranking across the seven different tasks, where we computed the pairwise Spearman's ρ computed over model performance on each pair of tasks. The results are presented in Fig. 1. We observed moderate-to-high correlations between different tasks. Notably, performance on *FAU-AIBO*^{5} correlated more with *MSP-Podcast-v1.11* (.714) than with *FAU-AIBO*^{2} (.580), despite the fact that the latter two tasks share identical data. This indicates that there is no 'one-winner-takes-all' model. Rather, as is consistent with the 'no-free-lunch' theorem [48], a different model can be more appropriate for a given dataset and task.

Do newer/larger models perform better? Table II displays the Spearman's ρ between model performance and the year each model was created, its multiply-addition computations (MACs), and its number of parameters. We observed very low correlations overall, with the highest correlation between complexity (measured in MACs) achieved for *FAU-AIBO*^{5} being a meagre .230. We acknowledge that these correlations heavily depend on our particular selection of models. We account for this by also showing 95% bootstrap confidence intervals (CIs). The large variance in our results, ranging from low negative correlations ($> -.3$) to moderately high correlations ($> .3$), does not allow for robust conclusions.

TABLE I: UAR results for all speech-based models in the **exploration phase** using the hyperparameters defined above (*Adam*, 0.0001, 4/16) for both the 2- and the 5-class tasks of the 2009 INTERSPEECH Emotion Challenge using the *FAU-AIBO* dataset, as well as the 4-class categorical SER task of *MSP-Podcast-v1.11*.

	FAU-AIBO^2	*FAU-AIBO*^{5}	*MSP-Podcast-v1.11*
IS09-s-mlp	.620	.294	.250
IS09-d-lstm	.670	.347	.469
IS10-s-mlp	.670	.348	.504
IS10-d-lstm	.689	.406	.486
IS11-s-mlp	.499	.201	.502
IS11-d-lstm	.670	.373	.499
IS12-s-mlp	.501	.201	.501
AlexNet	.503	.200	.250
IS12-d-lstm	.679	.373	.496
IS13-s-mlp	.498	.201	.503
IS13-d-lstm	.664	.378	.505
eGeMAPS-s-mlp	.618	.245	.479
eGeMAPS-d-lstm	.667	.397	.491
Resnet50	.688	.428	.510
VGG^{19}	.499	.204	.250
IS16-s-mlp	.500	.201	.497
VGG^{16}	.646	.404	.471
IS16-d-lstm	.655	.382	.498
VGG^{13}	.665	.385	.250
VGG^{11}	.666	.200	.459
CRNN^{18}	.680	.392	.514
EfficientNet	.669	.353	.537
CRNN^{19}	.683	.372	.503
w2v2-L	.500	.200	.250
w2v2-b	.500	.200	.250
ConvNeXt^t	.500	.400	.518
ConvNeXt^l	.663	.409	.518
ConvNeXt^b	.665	.412	.513
ConvNeXt^s	.674	.406	.531
ETDNN	.678	.404	.528
CNN14	.692	.394	.524
hubert-b	.500	.200	.250
Swin^b	.528	.200	.250
Swin^t	.530	.242	.269
AST	.535	.300	.396
w2v2-L-vox	.640	.402	.561
hubert-L	.667	.418	.570
Swin^s	.672	.306	.250
w2v2-L-robust	.684	.411	.555
Whisper^s	.656	.279	.405
w2v2-L-12-avd	.684	.411	.609
Whisper^t	.684	.380	.387
Whisper^b	.686	.200	.250

Agreement between different models: We first considered model agreement on *FAU-AIBO*. The average pairwise agreement (percentage of instances where two models agree) for all models of the exploration phase was 70% and 55% for the 2- and 5-class models of *FAU-AIBO*, which rises to 80% and 57%, respectively, when considering only the top-5 ones. Overall, this demonstrates that models trained on similar data do not converge to an identical solution – a finding congruent with the literature on underspecification [49]. Accordingly, the pairwise agreement for *MSP-Podcast-v1.11* was 55%/79% for all models and for the top-5 models respectively. These numbers indicate

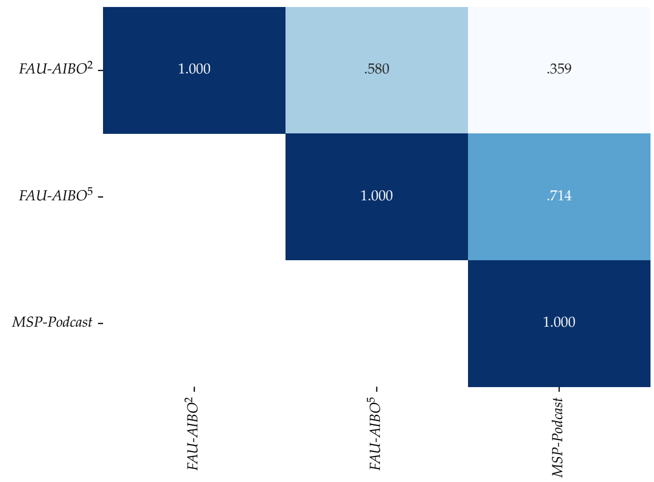


Fig. 1: Pairwise agreement in the relative ranking of models measured as Spearman's ρ between the UAR achieved by each model for two different tasks. A higher agreement indicates that models which perform well for the one task will perform better for the other task as well.

TABLE II: Spearman's ρ between model performance and year of publication, MACs, and number (#) of parameters of each model for each dataset and task. We additionally show 95% CIs obtained using bootstrapping (1000 random samples with replacement). Note that model MACs and parameters do not account for feature extraction.

	FAU-AIBO^2	*FAU-AIBO*^{5}	*MSP-Podcast-v1.11*
Year	.122 [-.204 ∣ .414]	.093 [-.201 ∣ .384]	.050 [-.291 ∣ .392]
MACs	.151 [-.157 ∣ .446]	.230 [-.087 ∣ .512]	.095 [-.263 ∣ .416]
# Parameters	-.083 [-.353 ∣ .195]	.085 [-.237 ∣ .393]	.026 [-.319 ∣ .356]

a moderate-to-high agreement on model predictions, especially for the best-performing models. This illustrates that, despite model size or year of publication, models trained on the same data tend to learn similar – but not identical – concepts.

Model vs human performance: We also investigated whether samples that are harder to classify for humans are also harder for models. The standard *FAU-AIBO* release comes with annotator confidences per instance, computed by taking the percentage of annotators who agree with the gold standard; we thus defined 'difficulty' as 1 minus that confidence. We then make the following observations when considering all models of the tuning phase:

a) We first adopted a model-agnostic measure of difficulty, which we defined as the number of models who disagree with the max-vote computed by all models on each instance – this is akin to the computation of difficulty for the human annotators. Spearman's ρ between this measure and annotator disagreement was moderate (.33 and .20 for the 2- and 5-class problems).

b) We then adopted a model-specific measure of difficulty, defined as the standard cross-entropy loss for each instance, similar to Hacohen and Weinshall [50] (see Rampp *et al.* [51] for our own investigations on the matter). Different models have different rankings of instance difficulty, with average pairwise ρ being .51 and .33 for the $2-$ and $5-$class problems.

TABLE III: UAR results for best-performing architectures in the *tuning phase* of *FAU-AIBO*. Showing best results obtained after tuning hyperparameters (batch size, optimiser, learning rate) and keeping the best-performing combination on the official test set. Also including 95% CIs for our models computed with bootstrapping. State-of-the-art results taken from original works.

Model	*FAU-AIBO*^{2}	*FAU-AIBO*^{5}
2009 Baseline	.677	.382
2009 Winners	.703	.417
2009 Fusion	.712	.440
Zhao *et al.* [24]	N/A	.454
IS10-d-lstm	.685 [.674 - .696]	.394 [.377 - .411]
Resnet50	.690 [.680 - .701]	.423 [.405 - .441]
CNN14	.672 [.661 - .683]	.448 [.428 - .467]
w2v2-L-12-avd	**.717 [.706 - .728]**	.448 [.431 - .465]
Whisper^t	.707 [.696 - .718]	**.454 [.437 - .472]**

TABLE IV: UAR results for best-performing architectures in the *tuning phase* of *MSP-Podcast-v1.11*. Showing best results obtained after tuning hyperparameters (batch size, optimiser, learning rate) and keeping the best-performing combination on the official test set. Also including 95% CIs for our models computed with bootstrapping.

Model	UAR
EfficientNet	.531 [.523 - .540]
hubert-L	.624 [.615 - .633]
w2v2-L-robust	.563 [.554 - .571]
w2v2-L-vox	.554 [.546 - .562]
w2v2-L-12-avd	.650 [.642 - .658]

c) Finally, we computed the Spearman's ρ between each model's UAR and its ρ with annotator disagreement; here, ρ was $-.38/-.07$ for the 2/5-class problem, indicating that larger agreement with human annotators did not lead to better performance (rather, the opposite).

We performed a similar analysis for *MSP-Podcast-v1.11*. Here, we did not directly have annotator confidence, as samples where annotators disagree have been filtered out by the original authors[4]. Instead, we adopted two of the dimensional annotations (*arousal* and *valence*) as proxies for the difficulty of a sample. Our intuition was that samples with a very high, or very low, arousal and/or valence, would be easier to distinguish than others. Our model-agnostic measure of difficulty (percentage of models which agree with the max-vote) had a Spearman's ρ of $-.15$ with arousal and $-.07$ with valence. The average Spearman's ρ of the instance-level loss of each model with arousal was .17 and with valence $-.18$

Collectively, our results indicate that models appear to learn differently than humans, with small agreement to what constitutes an easy or hard example. This was true for both *FAU-AIBO* and *MSP-Podcast-v1.11*. This points to the fact that, on the one hand, there is still room for further growth, but also that models may fail in ways that are considered 'catastrophic' by human observers.

[4]These were marked as "no agreement" in the original data and were excluded by filtering for the 4 classes we use here.

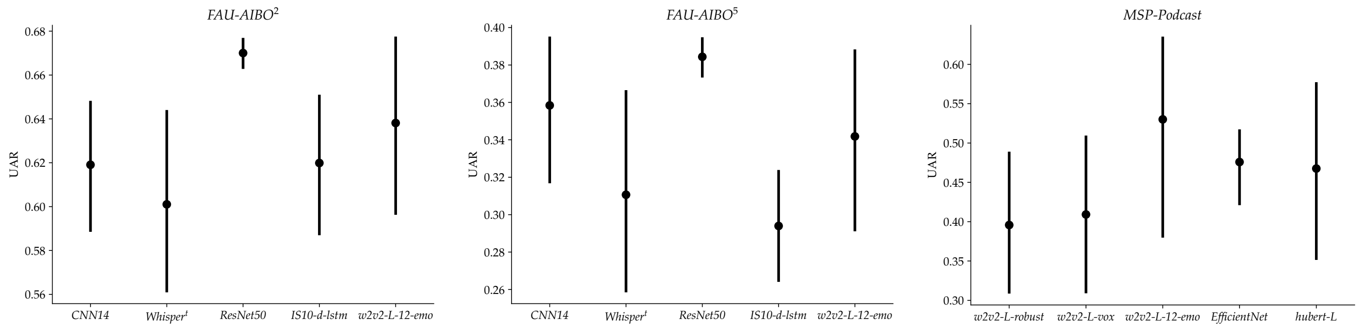


Fig. 2: Results of the *tuning phase* for *FAU-AIBO*^{2} (left), *FAU-AIBO*^{5} (middle), and *MSP-Podcast-v1.11* (right). Showing distribution of UAR values for the different hyperparameters tested for the top-5 performing models in each task.

Tuning phase: Turning to the *tuning phase* of *FAU-AIBO* and *MSP-Podcast-v1.11*, Table III and Table IV show the *best* performance for the top five models from Table I after optimising standard hyperparameters (optimiser, learning rate, batch size). For *FAU-AIBO*, *w2v2-L-12-avd* yielded the best performance for the 2-class problem of *FAU-AIBO* (.717) and *Whisper*^{t} for its 5-class problem (.454) – in both cases, we reached results better than the challenge winners, albeit with the gains for the 2-class problem being marginal. Given that *Whisper* has been trained for multilingual ASR (including German), and that performance improvements on valence prediction for English speech heavily depend on implicit linguistic knowledge (see Triantafyllopoulos *et al.* [12]), we expect *Whisper*'s success to be also attributed to that aspect (see Section IV-B for how text-based models performed). However, it is still the case that all models we tested remained close to or even below the original challenge baseline and winners, and especially the fusion of the top challenge submissions. This remained so even after selecting only the best-performing model out of all tested hyperparameters, essentially following a generally erroneous practice of overfitting. This was done intentionally to gauge performance under the most optimistic of settings – that of virtually unrestricted evaluation runs. We note that the original challenge participants were given 25 runs each. This shows how the gains obtained here must be further tempered to account for more runs on our side.

For *MSP-Podcast-v1.11*, the best performing model was a variant of *w2v2-L-12-avd*, trained with *SGD*, a learning rate of .01, and a batch size of 4, which reached a UAR of .650 $[.642 - .658]$ on the test set. We release this model, which we hence refer to as *w2v2-L-12-emo* publicly[5]. The performance for other models also improved, with the best hyperparameter combination for *hubert-L* reaching $.624[.615 - .633]$, illustrating once more how hyperparameter tuning can make a substantial difference in performance comparisons.

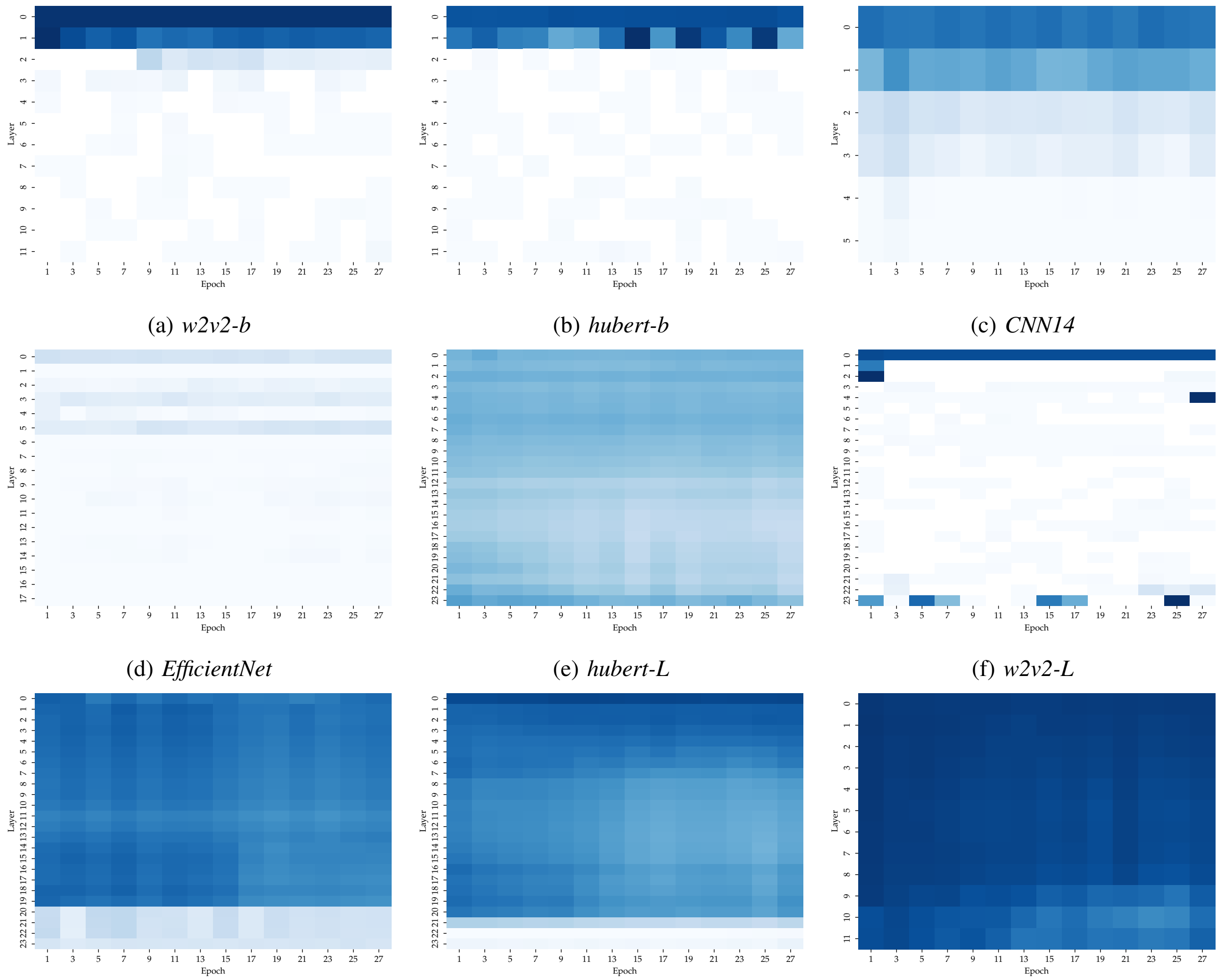
Fig. 2 shows all tuning results on both *FAU-AIBO* tasks and *MSP-Podcast-v1.11*. We observed a very high variability when changing hyperparameters – a well-known phenomenon impacting DL [52]. On the positive side, this tuning allowed us to obtain much higher performance – on *MSP-Podcast-v1.11* this gave rise to a UAR of .650 for *w2v2-L-12-avd*. However, the high variability calls attention to the fact that a different choice of hyperparameters in the exploration phase could have resulted in both different overall performance and, crucially, a different ranking across architectures. Unfortunately this is a problem that plagues all DL research at the moment, with the only possible solution being the investment in more compute power to explore a wider space of hyperparameters (which was not possible in our case).

2) OOD results: OOD results are shown in Table V. In many cases, OOD UAR is, surprisingly, higher than IID. We interpret this a side-effect of the datasets being much 'easier' than *MSP-Podcast-v1.11*, as they only contain acted, and very prototypical, data. Nevertheless, it is a promising sign of model generalisation. The best OOD performance in both cases was achieved by *w2v2-L-12-avd*, which reached a UAR of .806 on *EmoDB* and .617 on *IEMOCAP*. The Spearman's ρ of IID with OOD performance was .909 for *EmoDB* and .843 for *IEMOCAP*. This is another positive finding as it entails that selecting the best-performing model based on IID performance alone will translate to the best-performing model on OOD as well. However, the Spearman's ρ between year, MACs, or number of parameters and OOD UAR remained low for both *EmoDB* and *IEMOCAP* (year: $r = .112/.098$; MACs: $r = .166/.064$; # parameters: $.172/.096$).

3) Transfer learning dynamics of categorical SER models: We next considered the transfer learning dynamics of categorical SER models trained on *MSP-Podcast-v1.11*, as this is the largest dataset we have trained on. We computed the centred kernel alignment (CKA) [53], a measure of similarity for hidden representations using *EmoDB* as a probing dataset due to its smaller size. We did so for all transformer models, namely, *w2v2-b*, *w2v2-L*, *w2v2-L-vox*, *w2v2-L-robust*, *w2v2-L-12-avd*, *hubert-b*, and *hubert-L*, as well as two exemplary CNNs, *CNN14* and *EfficientNet*. For each model, we computed the CKA between hidden representations extracted from the initial state before fine-tuning, and all subsequent intermediate states which were saved every second epoch. For transformers, we extracted the representations after each hidden transformer block, for *CNN14* we did so after each of its convolutional blocks, and similarly for *EfficientNet*.

Results are shown in Fig. 3. Broadly, we note that all models exhibited a pattern of changing bottom-first (closer to the output), consistent both with the intuition that earlier layers

[5]https://huggingface.co/autrainer/msp-podcast-emo-class-big4-w2v2-l-emo



(a) *w2v2-b* (b) *hubert-b* (c) *CNN14*

(d) *EfficientNet* (e) *hubert-L* (f) *w2v2-L*

(g) *w2v2-L-robust* (h) *w2v2-L-vox* (i) *w2v2-L-12-avd*

Fig. 3: Transfer learning dynamics of models fine-tuned on 4-class SER using *MSP-Podcast-v1.11*. Plots show CKA between representations extracted at the initial state and at intermediate checkpoints stored every second epoch. A higher CKA (darker colour) corresponds to higher similarity between the two states, i. e. , a higher amount of feature reuse.

learnt more general, and thus re-usable features [54], and prior results [55, 56]. However, a closer look at individual models reveals important differences. Transformers that saw more adaptation of earlier layers, like *w2v2-b*, *hubert-b*, and *w2v2-L*, performed at chance-level performance. This is an indication that deviating a lot from the pretrained state can have catastrophic effects. We hypothesise that this was a side-effect of using a learning rate that was too high for these models. Interestingly, a similar pattern was exhibited by *EfficientNet*; however, this model still achieved good SER performance. We hypothesise that this particular model needed to adjust its representations more as it had been only pretrained on *ImageNet* data. Lastly, *w2v2-L-12-avd* showed the least adaptation overall, which is expected as the model had already been trained on *MSP-Podcast-v1.7* in a previous step (for dimensional SER) and had thus already adapted to the distribution of the data.

4) Probing transformer representations: Results are shown in Table VI. We note that better-performing models showed a higher Pearson's r overall across all features, indicating that representations that contained more information about those features led to a better SER performance. Comparing errors across features also uncovers interesting insights. For instance, μ(P) had a much higher Pearson's r than σ(P), which shows that transformers were better at grasping average pitch rather than pitch variations as also shown by the high errors on jitter. Given the importance of pitch variations for SER [57], this might present an avenue for future improvements.

5) Robustness to additive noise: The Spearman's ρ between the UAR achieved for clean and noisy audio was $.953$. However, as before, the Spearman's ρ between noisy UAR and year of publication ($.05$), MACs ($.10$), and # of parameters ($.03$) was extremely low. Thus, while IID performance showed a strong correlation with robustness, this robustness did not increase based on the year of publication or model size. Detailed noise robustness results are shown in Appendix D where we observe substantial degradations of model performance even for the newest models, with the best-performing *w2v2-L-12-avd* dropping from a UAR of $.609$ to one of $.546$.

TABLE V: OOD UAR results for the 4-class categorical SER models trained on *MSP-Podcast-v1.11* when evaluated on *EmoDB* and *IEMOCAP*.

	EmoDB	*IEMOCAP*
IS09-s-mlp	.250	.250
IS09-d-lstm	.487	.339
IS10-d-lstm	.423	.354
IS10-s-mlp	.553	.489
IS11-d-lstm	.440	.373
IS11-s-mlp	.630	.503
AlexNet	.250	.250
IS12-d-lstm	.504	.315
IS12-s-mlp	.598	.489
IS13-s-mlp	.627	.495
IS13-d-lstm	.489	.365
eGeMAPS-s-mlp	.530	.491
eGeMAPS-d-lstm	.361	.382
Resnet50	.691	.470
VGG^{13}	.250	.250
VGG^{19}	.250	.250
VGG^{11}	.549	.409
VGG^{16}	.596	.490
IS16-s-lstm	.649	.466
IS16-d-mlp	.460	.351
CRNN^{18}	.648	.445
CRNN^{19}	.642	.519
EfficientNet	.590	.406
w2v2-L	.250	.250
w2v2-b	.250	.250
ConvNeXt^{b}	.653	.447
ConvNeXt^{l}	.684	.505
ConvNeXt^{t}	.705	.510
CNN14	.715	.429
ETDNN	.600	.501
ConvNeXt^{s}	.669	.519
hubert-b	.250	.250
Swin^{b}	.250	.250
Swin^{s}	.250	.250
Swin^{t}	.250	.320
AST	.505	.431
w2v2-L-robust	.771	.499
w2v2-L-vox	.783	.526
hubert-L	.756	.544
Whisper^{b}	.250	.250
Whisper^{t}	.401	.377
Whisper^{s}	.503	.381
w2v2-L-12-avd	.806	.617

6) Individual fairness: We finish this section with a discussion of speaker-level performance for audio-based models. To do so, we computed the speaker-level performance for each task and used that as the utility to compute the Gini index.

We are interested in two main questions: a) what was the average *equality* observed for a dataset across models, as measured by the Gini index, and, b) were models that performed better on the instance-level also more fair across speakers, i. e., what was the Spearman's ρ between the UAR of a model on the instance-level and its Gini index across speakers? We only considered models that performed better than chance, defined as a UAR larger than .01 more than the corresponding random-chance UAR, i. e., .51 for 2-class, .26 for 4-class, and .21 for 5−class, respectively. Our results are presented in Table VII.

We note that the only task which showed a moderate average inequality across speakers is *MSP-Podcast-v1.11*, with an average Gini index of .329. On the positive side, it also showed a high *negative* correlation between instance-level UAR and speaker-level Gini index ($-.344$). This means that models which performed better overall across all instances were also *more fair* towards different speakers. The *FAU-AIBO* tasks exhibited a low to moderate average Gini index and a negligible correlation between instance-level UAR and the Gini index across speakers. Based on this, we conclude that speaker inequality affects different datasets and tasks to different degrees, but it remains an open avenue for future research considering the potential of *personalisation* methods to improve fairness and overall performance [45].

The Spearman's ρ between the Gini index and year of publication ($-.04$), MACs ($.-25$), and # of parameters ($-.07$) for *MSP-Podcast-v1.11* was either zero or mildly negative showing that bigger models were slightly more fair than smaller ones. However, this trend is contradicted by results for *FAU-AIBO*^{2} ($.29/.55/.53$) and *FAU-AIBO*^{5} ($.18/.32/.26$) which showed moderate-to-strong Spearman's ρ between the Gini index and year of publications, MACs, and # of parameters, pointing towards *lower individual fairness* for newer and bigger models.

B. Text-based models

In addition, we used state-of-the-art text-based models to predict emotion on both *FAU-AIBO* tasks and *MSP-Podcast-v1.11*. UAR results are shown in Table VIII.

The highest performance for *FAU-AIBO*^{2} was achieved by *DistilBERT* at .667, for *FAU-AIBO*^{5} by *Llama-3* at .401, and for *MSP-Podcast-v1.11* by *Llama-2* at .564, which was essentially on-par with *Llama-3* at .563 and *Mistral* at .560. Notably, all these results were lower than the best-performing audio-based models though competitive with models trained on the exploration phase. For instance, *Llama-2* results on *MSP-Podcast-v1.11* would rank third in our *exploration phase*, trailing marginally behind *hubert-L* which scored at .570. This demonstrates how text contains vital information for the SER task of *MSP-Podcast-v1.11*.

Results on *FAU-AIBO* were overall lower, with text-based models scoring in the middle of the 'competition'. We hypothesise that this is a side-effect of the fact that the textual content in *FAU-AIBO* is much more limited than in *MSP-Podcast-v1.11* given the Wizard-of-Oz scenario in which it was recorded. Nevertheless, the fact that text-based models obtained better-than-chance performance is in itself remarkable, given how limited the textual content is, and indicates that there are potential systematic biases with respect to the linguistic content of certain emotions. We discuss the linguistic content of both datasets further in Appendix E.

1) Complementarity between audio & text: We finally considered the complementarity between audio-based and text-based models. In the present subsection, we rely on error analysis to investigate whether text-based models, which generally underperform audio-based ones, brought additional benefits for classification or merely predicted correctly a subset of the data where audio-based models already performed well.

For *MSP-Podcast-v1.11* we compared two exemplary audio-based models from the exploration phase, *w2v2-L-12-avd*, as the best-performing transformer, and *EfficientNet*, as the best-performing non-transformer model, and the best-performing

TABLE VI: Average Pearson's r over all layers and epochs when probing transformer representations for their knowledge of acoustic features using linear probes on *EmoDB*. A higher Pearson's r – either positive or negative – reflects the strength with which this feature is encoded in internal representations, and how it might be possibly utilised during inference. The last column shows UAR taken from Table I.

Model	μ(P)	σ(P)	μ(L)	μ(L) [dB]	μ(J)	μ(S)	μ(HNR)	μ(F1)	μ(F2)	μ(F3)	UAR
w2v2-b	.863	.172	.454	.271	.185	.318	.519	.603	.458	.313	.250
hubert-b	.523	.149	.304	.207	.055	.114	.250	.380	.306	.228	.250
w2v2-L	.547	.187	.292	.200	.060	.142	.261	.365	.286	.217	.250
w2v2-L-robust	.926	.088	.361	.297	.112	.247	.518	.726	.597	.409	.555
w2v2-L-vox	.941	.213	.504	.435	.175	.275	.584	.772	.653	.473	.561
hubert-L	.944	.220	.451	.382	.100	.272	.547	.781	.669	.471	.570
w2v2-L-12-avd	.959	.237	.533	.420	.188	.350	.649	.815	.697	.521	.609

TABLE VII: Average Gini index of speaker-level performance for all models and Spearman's ρ between that Gini index and global performance (UAR). A higher μ(Gini) indicates more inequality across speakers for all models trained on a given task. A lower ρ shows that models with the highest performance have a lower inequality as well.

Task	ρ	μ(Gini)
FAU-AIBO^2	.075	.078
FAU-AIBO^5	-.015	.139
MSP-Podcast-v1.11	-.344	.329

TABLE VIII: UAR results for text-based models on *FAU-AIBO* and *MSP-Podcast-v1.11*.

Model	*FAU-AIBO*^{2}	*FAU-AIBO*^{5}	*MSP-Podcast-v1.11*
BERT	.647	.393	.528
RoBERTa	.646	.381	.541
DistilBERT	**.667**	.391	.523
Electra	.666	.386	.536
Llama-2	.662	.389	**.564**
Llama-3	.645	**.401**	.563
Mistral	.623	.378	.560

text-based model, *Llama-2*. We also considered *FAU-AIBO*^{2} using *w2v2-L-12-avd*, *CNN14*, and *DistilBERT* as the three models showing a more balanced performance over *FAU-AIBO*^{2} and *FAU-AIBO*^{5}.

We first examined the pairwise agreement between all model combinations, measured as the percentage of samples where two models make the same decision. For *MSP-Podcast-v1.11*, this lied at .556 between *Llama-2* and *w2v2*, .451 between *Llama-2* and *EfficientNet*, and .639 between *EfficientNet* and *w2v2*. For *FAU-AIBO*^{2}, we also observed a larger agreement between *w2v2-L-12-avd* and *CNN14* (.815) than between *w2v2-L-12-avd* and *DistilBERT* (.666) or *CNN14* and *DistilBERT* (.672).

Additionally, we measured their complementarity by comparing the UAR of each model on the entire test set (*Test-1*) vs its UAR on the subset of the test set that each other model classifies correctly (+; green) or incorrectly (−; red). These results are shown in Section IV-B1. On *MSP-Podcast-v1.11*, *w2v2-L-12-avd* showed higher UAR when either *Llama-2* or *EfficientNet* performed well – a sign that its correct predictions were a superset of the correct predictions of the other two models. In contrast, *EfficientNet* and *Llama-2* only showed a performance improvement when evaluated on the samples that *w2v2-L-12-avd* classified correctly, thus showing little overlap between these pairs of models.

These results lead us to conclude that the complementarity was much higher between *EfficientNet* and *Llama-2*, than between *w2v2-L-12-avd* and any of the other two models. We hypothesise that this happens because *w2v2-L-12-avd* already captured linguistics implicitly, as shown in Triantafyllopoulos *et al.* [12], whereas *EfficientNet* did not. Seen in a different light, this means that *w2v2-L-12-avd* – and, generally, other pretrained transformers – seem capable of combining 'the best of both worlds'.

Overall, this investigation revealed that audio-based models agreed more with one another than they agreed with a text-based model. However, the differences for *FAU-AIBO* were less pronounced than for *MSP-Podcast-v1.11*. We hypothesise that this was the case because a) *FAU-AIBO* is a German dataset and *w2v2-L-12-avd* had not been pre-trained on German data, and b) *FAU-AIBO* was collected in a Wizard-of-Oz scenario where linguistics play a minor role in the expression of emotion; hence, text-based models performed worse and audio-based models could draw little additional information from linguistics.

V. Discussion & Conclusion

In this section, we jointly consider all findings presented in the previous section and connect them to related work.

We begin with the most important research question that we set out to answer: is recent progress on SER obtained by using newer and bigger DL models monotonic? Our answer was inconclusive. Bigger and newer models did not necessarily lead to better IID or OOD performance, more robustness, or more fairness. In fact, we also observed opposing trends with newer/bigger models often showing lower fairness or worse performance. However, one notable outcome from our work is that any conclusions regarding progress are pre-conditioned on the particular set of models that are evaluated. As shown in Section IV-A, varying the set of models included in the analysis can have dramatic effects on observed correlation between model performance and year or computational complexity. This has major implications for any attempts to gauge progress on a long-term horizon. Replication studies like ours are inevitably

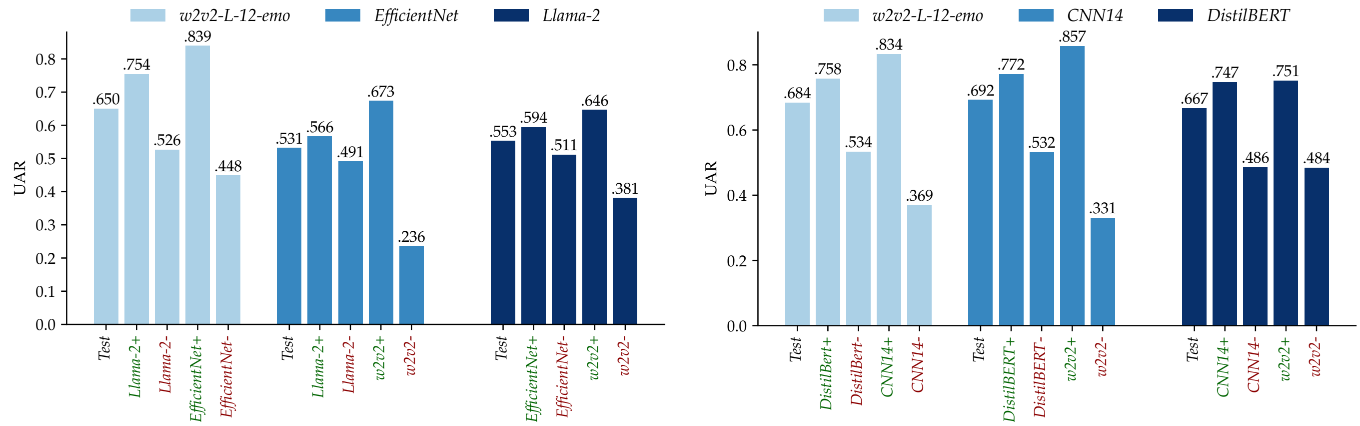


Fig. 4: UAR for each model on *MSP-Podcast-v1.11* (left) and *FAU-AIBO*[2] (right) on the entire test set vs on the subset that each of the three models examined here classifies correctly (+; green) or incorrectly (−; red).

limited by the effort we may put in them and our subjective biases when selecting the methods that are included. Seen in a different light, reporting CIs on measures of progress is vital to avoid a misrepresentation of the underlying findings.

The most important caveat underlying our work is the strong impact of hyperparameters on model performance, as showcased in our *tuning* phase. The large variability obtained for the models which were optimised during the tuning phase indicates that it is hard to identify a clear winner unless a sufficient amount of variations are tested. This point is further discussed as a limitation of our work in the next section. However, we note that similar computational constraints plague most recent AI works, highlighting, once again, the need for exploring a space of hyperparameters as wide as possible – and certainly giving compared models a 'fair chance' by considering at least a few different hyperparameters and reporting the range of results.

Moreover, our work can be positioned within the context of the *scaling hypothesis*, which is currently driving the debate in the broader AI community [58] [6]. Is the solution to the SER problem merely a matter of scaling to bigger models and bigger data? We have shown that bigger models do not necessarily lead to better performance, though this was only done when comparing *across* architectures; the scaling hypothesis is typically applied within the same architecture family (add more and wider layers). Our computational resources did not allow us to investigate this question which is certainly worth pursuing in future work. However, understanding the inner workings of more successful models can also help pave future advances independent of scaling. For instance, our probing analysis revealed that state-of-the-art transformer models still struggle with predicting pitch variability. Designing novel architectures that are able to do so, or pre-training tasks that lead to models doing so, might be another interesting avenue for future research, especially in the face of scarce computational resources.

Ultimately, the scaling hypothesis is an important and pressing question for our field that our work has not conclusively answered. While our results point against it, we have not nearly reached the levels of scale commonly achieved by "frontier models" (on the scale of GPT-4o). In the case of academic research, this amount of compute resources is potentially unattainable in the near-term future. At the very least, the recent stabilisation of *MSP-Podcast*, and the existence of streamlined training pipelines, like our *autrainer* toolkit [35], represent an opportunity to establish a common benchmark for naturalistic SER that will allow comparisons of long-term progress.

VI. Limitations

We are aware that our work suffers from certain limitations, which we aim to discuss here.

Insufficient data – From the multitude of available SER datasets, we chose to focus on only two, with agreement in the ranking of model performance across datasets being moderate-to-high. See Appendix A for a longer discussion on why we chose these two particular datasets.

Insufficient models – With hundreds of DL papers published in the domain of SER in recent years, it was impossible to replicate all of them. We opted to focus on the ones that were easier to replicate from a coding perspective. Our most important omission was the non-inclusion of large audio-language models [59, 60], i. e., models which connect an audio encoder to an LLM decoder for joint audio-text modelling. While extremely promising, there were not many models available at the time when we conducted our experiments.

Insufficient tuning – Our investigation of hyperparameter tuning revealed that model performance heavily depends on the choice of parameters. Scaling up that search is contingent upon the availability of compute power. It could very well be that our choice of hyperparameters for

[6]Note that this thread of research is commonly referred to as "neural scaling laws". Given that these "laws" are merely experimental observations, rather than mathematical certainties, we prefer the softer term "hypothesis".

the exploration phase may have favoured some models over others.

Inconclusive outcomes – The dependence of our results on the particular selection of models and the large impact of hyperparameters highlights the uncertainty of our findings. Nevertheless, counter to overall consensus, we did not observe a clear and consistent trend favouring bigger and more recent models. Furthermore, any observed gains can be attributed to a selection of hyperparameters that favoured some models more than others.

We acknowledge all of the above limitations. They are all known problems that plague the entire AI community. In fact, our work can be seen in direct connection to works highlighting issues of *comparability* and *replicability of findings* [61, 62, 63, 64]. Our biggest takeaway is to interpret recent findings with the same care; hyperparameter tuning, rather than architectural innovation, may be responsible for reported empirical gains. In particular, we urge for caution with regards to the *scaling hypothesis*, i. e., the hypothesis that bigger models trained on more data leads to better results. The same limitations that plague our work are also important for works that provide confirmatory evidence for it – perhaps even more so given the possibility of publication bias in favour of positive results. As confirmation or refutation of this hypothesis stands to be a defining model for our field, we hope that our investigation serves as a reminder that the replication of *findings* is oftentimes just as important as the replication of *results*.

VII. Future work

We end our contribution by considering potential pathways for future research. A major bottleneck in such large-scale replication studies is the lack of compute capacity. For academic institutions with modest computational resources this constraint can only be overcome by intense cross-institutional collaboration; resources can be pulled to increase the number of investigated models and hyperparameters. Shared authorship for shared compute could be a model that facilitates such large-scale collaboration; perhaps we could envision scientific publications with longer lists of authors as those seen in other fields, like physics.

Beyond that, it is necessary to invest more effort in interpreting the ways in which models make their decisions (interpretability/explainability) and the ways in which they fail (error analysis/fairness/robustness). These analyses can uncover the weak-points of contemporary architectures and inspire novel ways to mitigate them. For instance, the fact that transformer models seem to be worse at predicting pitch variations than pitch itself is a clue worth further investigation.

Finally, it is perhaps time to begin considering what may lie beyond deep learning [65]. Perhaps we have reaped the benefits of incorporating feature extraction with classification and learning both from data (the so-called *end-to-end* paradigm), and further increasing the complexity of models and the data they ingest can only offer diminishing returns. Recent progress on *reasoning* models offers a promising avenue to integrate pre-trained 'world knowledge' with contextual features analysis to help overcome the inherent uncertainty in predicting human emotions in a naturalistic environment.

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VIII. References

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Appendix

A. Dataset selection

The selection process that resulted in the two datasets used for benchmarking, *FAU-AIBO* and *MSP-Podcast*, was far from trivial. In the present section, we provide a justification for our selection criteria.

Our first point of consideration was the *context* in which the portrayed emotions arose. Ultimately, SER systems that are deployed in real-world applications will inevitably have to process spontaneous human expressions that correspond to emotional triggers in the real world. Improving performance on acted data does not necessarily equate an equal improvement on naturalistic data. This led us to exclude most commonly-used datasets such as *IEMOCAP* [11], *RAVDESS* [19], *CREMA-D* [10], or *EmoDB* [15] for our in-domain training as they all comprise exclusively acted content[7].

We note that neither *FAU-AIBO* nor *MSP-Podcast* are perfect in this regard. *FAU-AIBO* was collected in a very artificial Wizard-of-Oz scenario and the (underage) subjects were aware that they were being recorded. As such, their portrayal of emotion might have been mediated by this 'observer effect'. However, using a Wizard is an established procedure in elicitation studies, with the underlying premise being that the elicited reactions are similar to the ones found in "in-the-wild" scenarios. On the other hand, *MSP-Podcast* is also not free from criticism in this regard, given that its material was sourced from podcasts which aired in public domains. Even though the subjects were (probably) not under the impression of being monitored, they were certainly aware that they were speaking in front of an audience. While this is a valid context in which to measure human emotions, it is *a* context nonetheless.

Eventually, we had to surrender to the fact that finding a dataset which encompasses the whole palette of human experience, with participants further being entirely unaware of their ever being monitored, would be impossible without resorting to unsolicited surveillance. For this reason, we settled for *FAU-AIBO* and *MSP-Podcast* – both encompassing only a specific snapshot of the human experience, which, limited though it may be, is still much wider than most datasets comprising solely acted emotional speech.

Our next desideratum was the *coding scheme*, where we distinguish between schemes driven by *theory*, and schemes driven by *observation*. The vast majority of SER datasets follow a prescriptive, top-down approach and use a set of pre-defined emotional categories. Usually, these correspond to – or at the very least are inspired from – Ekman's "big-6" [66], plus a "neutral" category to designate a complete lack of emotion. This process is inherently *model-based*. In other words, it enforces a specific, pre-defined model of human emotions on the annotation process. *MSP-Podcast* follows this recipe by adopting 8 categorical labels (plus 1 "other" class), which we further restrict to the four categories most commonly-found in SER literature {anger, happiness, neutral, sadness}.

While this enables a seamless translation of its annotations to many other SER corpora, it is still enforcing a particular model of human emotions and their expressions on the annotators. This model is not without criticism and might be too limited to capture the entire spectrum of human experience [67].

On the other end of the theory-observation divide are *data-driven* approaches. This is the paradigm that *FAU-AIBO* primarily adheres to. The set of candidate labels was developed iteratively by experienced annotators following a consensus approach. Then, the data was annotated using an original set of 11 labels on the word-level. Subsequently, a majority vote for the five annotators was taken over all word-level annotations in an utterance to derive an utterance-level annotation. The amount of labels was then reduced to 5 and 2 emotions, respectively, using a model-based mapping. As such, the dataset comprises a set of very unique emotion labels which do not easily translate to other contexts – but which, importantly, are tailored to the specific context that this dataset encapsulates.

The choice of coding scheme has important repercussions about the applicability of trained models and leads to a much larger debate. In colloquial terms, our overarching goal is to build algorithms that are able to *correctly classify* human emotions in *all* contexts and in *all* situations. Following generic, model-based approaches eases the translation of model predictions to new contexts, but to the detriment of specificity. On the other hand, building models that are tailor-made to specific scenarios suffers from scalability and risks the fragmentation of research efforts. The search for ways to accommodate these two paradigms is a very important research direction, but one that was beyond the scope of our work.

[7]Nevertheless, we do use two of those datasets, *IEMOCAP* and *EmoDB*, to gauge out-of-domain generalisation given the relative lack of naturalistic datasets.

Settling for one model-based and one data-driven dataset was a good compromise.

Finally, *dataset size* and *fame* were two last considerations. Given our focus on DNNs, we naturally favoured datasets with more training instances. Moreover, we aimed for datasets that are more widely-used by, or at least known to, the broader SER community. Among the ones that are not acted, *MSP-Podcast* and *FAU-AIBO* were two reasonable choices.

B. Models

This section contains the details for all pre-trained models used in our work.

Table IX records the configuration files used to extract *openSMILE* features. We accommodate both static and dynamic versions of the official feature sets. For the *dynamic* features, we used a 2-layered long short-term memory network (LSTM) model with 32 hidden units, followed by mean pooling over time, one hidden linear layer with 32 neurons and ReLU acivation, and one output linear layer; all hidden layers are followed by a dropout of 0.5; these models are denoted with $-d-lstm$. Additionally, we train 3-layered MLPs with 64 hidden units each, a dropout of 0.5, and ReLU activation for the *static* features; these models are denoted with $-s-mlp$.

Table X contains the paths to the model definitions of CRNNs models. We used two particular instantiations introduced by Tzirakis *et al.* [68] (*CRNN*^{18}) and Zhao *et al.* [69] (*CRNN*^{19}).

Table XI contains the paths to model code and pre-trained states used for spectrogram-based models that have been pre-trained on vision data. We used *AlexNet*, *Resnet50*, all versions of *VGG* (^{11,13,16,19}), EfficientNet-B0 (*EfficientNet*), the tiny, small, base, and large versions of *ConvNeXt* (^{t,b,s,l}), and the tiny, base, and small versions of the *Swin* Transformer (^{t,b,s}). In all cases, we use the best-performing model state on *ImageNet* as available in the *torchvision-v0.16.0* package. As features, we always used the Mel-spectrograms generated for *CNN14*, i. e., 64 Mels with a window size of 32 ms and a hop size of 10 ms; the resulting matrices were then replicated over the three dimensions to generate the 3-channel input that is required by models designed for computer vision tasks.

Table XII lists the models pre-trained on supervised audio tasks. We used *CNN14* and *AST* pre-trained on *AudioSet*, as well as the *ETDNN* model pre-trained for speaker identification [27]. Furthermore, we fine-tuned *Whisper*, albeit only its three smallest available variants of (^{t,b,s}) due to hardware constraints.

Finally, Table XIII documents the paths of SSL models, which are largely the same as the ones used in Wagner *et al.* [7]. In this work, we used the pretrained states from the *base* and *large* variants of *w2v2* and *hubert* (*w2v2-b*, *w2v2-L*, *hubert-b*, *hubert-L*), a multilingual model trained on *VoxPopuli* [70] (*w2v2-L-vox*), a *'robust'* version of *w2v2* trained on more data [71] (*w2v2-L-robust*), as well as the pruned version of that model further finetuned for dimensional SER on *MSP-Podcast* that was presented in Wagner *et al.* [7] (*w2v2-L-12-avd*). Similar to Wagner *et al.* [7], we add an output $2-$layered MLP which takes the pooled hidden embeddings of the last layer as input.

Our text-based models are listed in Table XIV. While we use the same acronym to refer to both English and German version of some models, these do not always correspond to the same model state as some models were trained for a single language. We only use the "second-generation" LLMs for both languages as they were trained on multilingual data.

C. Codecs

We investigated the robustness of *w2v2-L-12-emo* on the following codecs:

MP3 (MPEG-1/2 Layer-3) combines psychoacoustic modeling and compression of the modified discrete cosine transform [72]. We used the *LAME* version available in *ffmpeg* [8]. Its psychoacoustic model was tuned on manual listening tests and focuses on maintaining the quality of specific frequency bands.

AAC (Advanced Audio Codec) was designed as a follow-up to *MP3* [72]. It improves on it by – among ohters – increasing the frequency resolution.

Opus combines a short- and a long-term (linear) predictor.

EnCodec is one of the most recent, data-driven codecs relying on DNNs [73]. It is a simple, CNN-based auto-encoder with an intermediate steps of residual vector quantisation. It has been trained on a large dataset comprising multiple audio sources.

SemantiCodec aims for higher compression rates by incorporating a very powerful encoder (*AudioMAE*) and a more powerful denoising diffusion probabilistic model (DDPM) decoder [74]. However, the improvement in compression rates comes with a substantial penalty on computational overhead.

D. Robustness to additive noise

Table XV presents detailed robustness results for all models of our exploration phase.

E. Linguistic content

Fig. 5 shows the word distribution for *FAU-AIBO* and *MSP-Podcast*[9]. We observe substantially higher variability for *MSP-Podcast*, which is consistent with its in-the-wild procurement. *FAU-AIBO*, on the other hand, comprises a much more constricted, intent-oriented vocabulary, featuring mostly simply commands ("geh nach links"), as well as the frequent repetition of the word *Aibo*.

[8] `https://lame.sourceforge.io/`

[9] Computed using the default parameters of the Python *WordCloud* package (`https://amueller.github.io/word_cloud/`).

TABLE IX: Feature URLs for *openSMILE*.

Features	URL
IS09-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/is09-13/IS09_emotion.conf`
IS10-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/is09-13/IS10_paraling_compat.conf`
IS11-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/is09-13/IS11_speaker_state.conf`
IS12-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/is09-13/IS12_speaker_trait_compat.conf`
IS13-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/is09-13/IS13_ComParE.conf`
IS16-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/compare16/ComParE_2016.conf`
eGeMAPS-s,d	`https://github.com/audeering/opensmile/blob/v3.0.2/config/egemaps/v02/eGeMAPSv02.conf`

TABLE X: Model URLs for end-to-end CRNNs.

Model	URL
CRNN^{18}	`https://github.com/end2you/end2you/blob/master/end2you/models/audio/emo18.py`
CRNN^{19}	`https://github.com/end2you/end2you/blob/master/end2you/models/audio/zhao19.py`

TABLE XI: Model URLs for computer vision models applied on spectrograms.

Model	URL
AlexNet	`https://pytorch.org/vision/0.16/models/alexnet.html`
Resnet50	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.resnet50.html#torchvision.models.resnet50`
VGG^{11}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.vgg11.html#torchvision.models.vgg11`
VGG^{13}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.vgg13.html#torchvision.models.vgg13`
VGG^{16}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.vgg16.html#torchvision.models.vgg16`
VGG^{19}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.vgg19.html#torchvision.models.vgg19`
ConvNeXt^{t}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.convnext_tiny.html#torchvision.models.convnext_tiny`
ConvNeXt^{s}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.convnext_small.html#torchvision.models.convnext_small`
ConvNeXt^{b}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.convnext_base.html#torchvision.models.convnext_base`
ConvNeXt^{l}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.convnext_large.html#torchvision.models.convnext_large`
Swin^{t}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.swin_t.html#torchvision.models.swin_t`
Swin^{s}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.swin_s.html#torchvision.models.swin_s`
Swin^{b}	`https://pytorch.org/vision/0.16/models/generated/torchvision.models.swin_b.html#torchvision.models.swin_b`
EfficientNet	`https://github.com/lukemelas/EfficientNet-PyTorch/blob/master/efficientnet_pytorch/model.py#L27`

TABLE XII: Model URLs for supervised audio models. *Whisper* models found at `https://huggingface.co/`.

Model	URL
CNN14	`https://github.com/qiuqiangkong/audioset_tagging_cnn/blob/master/pytorch/models.py`
ETDNN	`speechbrain/spkrec-ecapa-voxceleb`
Whisper^{t}	`openai/whisper-tiny`
Whisper^{s}	`openai/whisper-small`
Whisper^{b}	`openai/whisper-base`
AST	`MIT/ast-finetuned-audioset-10-10-0.4593`

TABLE XIII: Model URLs for self-supervised speech models (`https://huggingface.co/`).

Model	URL
hubert-b	`facebook/hubert-base-ls960`
hubert-L	`facebook/hubert-large-ll60k`
w2v2-b	`facebook/wav2vec2-base`
w2v2-L	`facebook/wav2vec2-large`
w2v2-L-vox	`facebook/wav2vec2-large-100k-voxpopuli`
w2v2-L-robust	`facebook/wav2vec2-large-robust`
w2v2-L-12-avd	`audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim`

TABLE XIV: Text-based models used for SER on *FAU-AIBO* and *MSP-Podcast-v1.11*. All models available at `https://huggingface.co/`.

Model	URL
	English models
BERT	`google-bert/bert-base-uncased`
RoBERTa	`FacebookAI/roberta-base`
DistilBERT	`distilbert/distilbert-base-uncased`
Electra	`google/electra-base-discriminator`
Llama-2	`meta-llama/Llama-2-7b-chat-hf`
Llama-3	`meta-llama/Meta-Llama-3-8B-Instruct`
Mistral	`mistralai/Mistral-7B-Instruct-v0.2`
	German models
BERT	`dbmdz/bert-base-german-uncased`
RoBERTa	`T-Systems-onsite/german-roberta-sentence-transformer-v2`
DistilBERT	`distilbert/distilbert-base-german-cased`
Electra	`german-nlp-group/electra-base-german-uncased`
Llama-2	`meta-llama/Llama-2-7b-chat-hf`
Llama-3	`meta-llama/Meta-Llama-3-8B-Instruct`
Mistral	`mistralai/Mistral-7B-Instruct-v0.2`

Fig. 5: Wordclouds visualising bi-gram occurrence frequency for *FAU-AIBO* (left) and *MSP-Podcast* (right).

TABLE XV: UAR on the original and noisy versions of the *MSP-Podcast-v1.11* test set. We use the models trained in the exploration phase of Section IV-A without additional retraining, i. e. , all models have only been trained on clean data.

Model	Original	Noisy
IS09-s-mlp	.250	.250
IS09-d-lstm	.469	.337
IS10-s-mlp	.504	.373
IS10-d-lstm	.486	.366
IS11-s-mlp	.502	.385
IS11-d-lstm	.499	.374
IS12-s-mlp	.501	.388
IS12-d-lstm	.496	.368
AlexNet	.250	.250
IS13-s-mlp	.503	.408
IS13-d-lstm	.505	.360
eGeMAPS-s-mlp	.479	.348
eGeMAPS-d-lstm	.491	.353
Resnet50	.510	.419
IS16-d-mlp	.498	.353
IS16-s-lstm	.497	.387
VGG^{11}	.459	.368
VGG^{13}	.250	.250
VGG^{16}	.471	.402
VGG^{19}	.250	.250
$CRNN^{18}$	.514	.419
EfficientNet	.537	.432
$CRNN^{19}$	.503	.406
w2v2-b	.250	.250
w2v2-L	.250	.250
$ConvNeXt^{t}$	.518	.435
$ConvNeXt^{s}$	.531	.449
$ConvNeXt^{b}$	.513	.427
$ConvNeXt^{l}$	.518	.423
ETDNN	.528	.450
CNN14	.524	.461
hubert-b	.250	.250
hubert-L	.570	.519
$Swin^{t}$	.269	.244
$Swin^{s}$	.250	.250
$Swin^{b}$	.250	.250
AST	.396	.323
w2v2-L-robust	.555	.430
w2v2-L-vox	.561	.431
w2v2-L-12-avd	.609	.546
$Whisper^{t}$	.387	.306
$Whisper^{s}$	.405	.302
$Whisper^{b}$	.250	.250